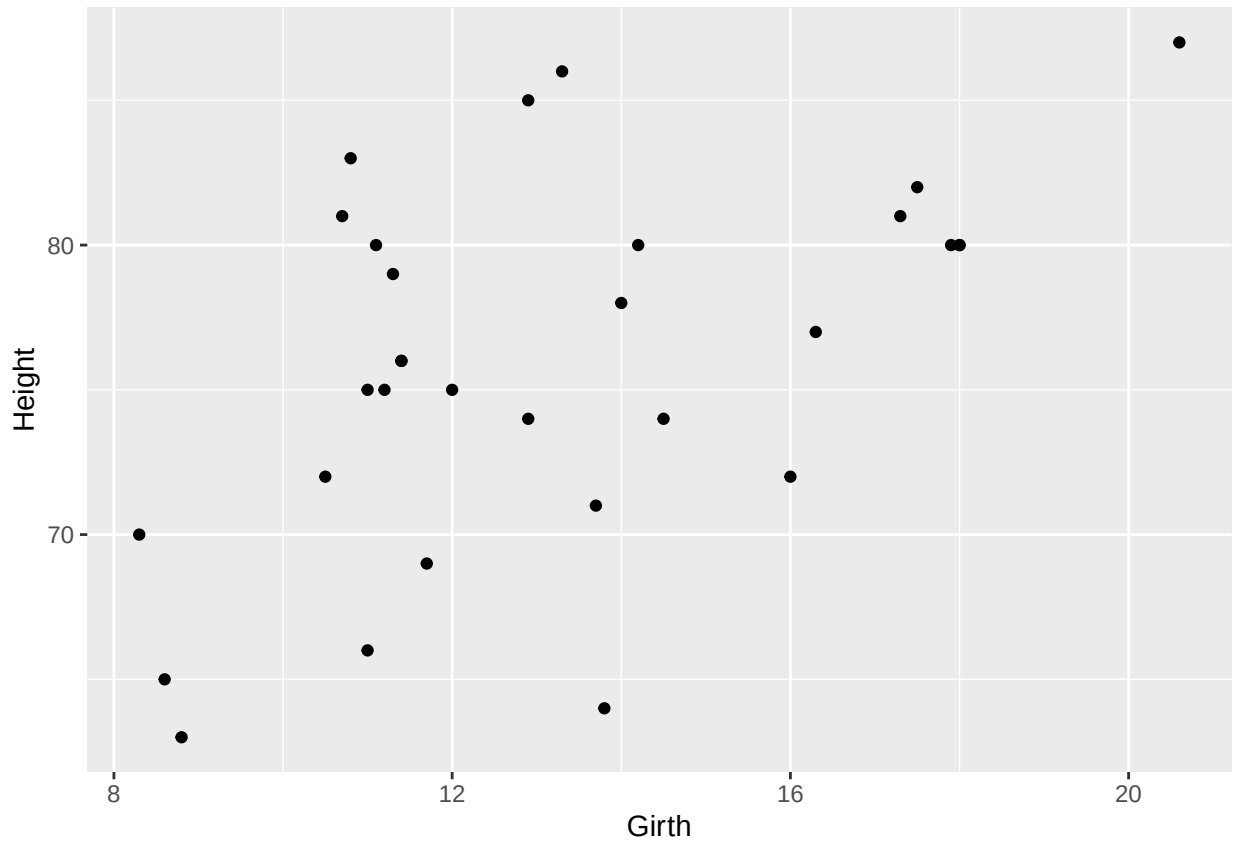


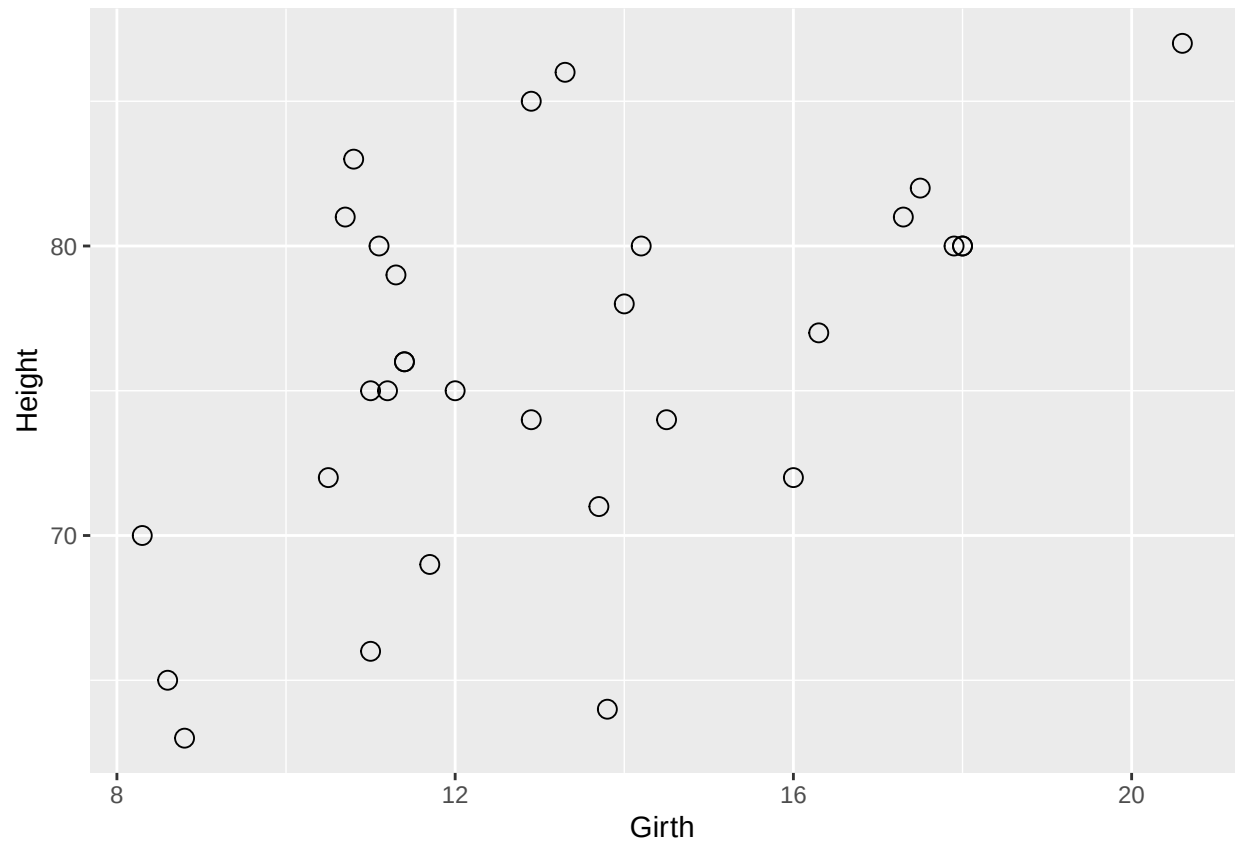
Scatter Plots

Noratiqah Mohd Ariff

Basic Scatter Plot

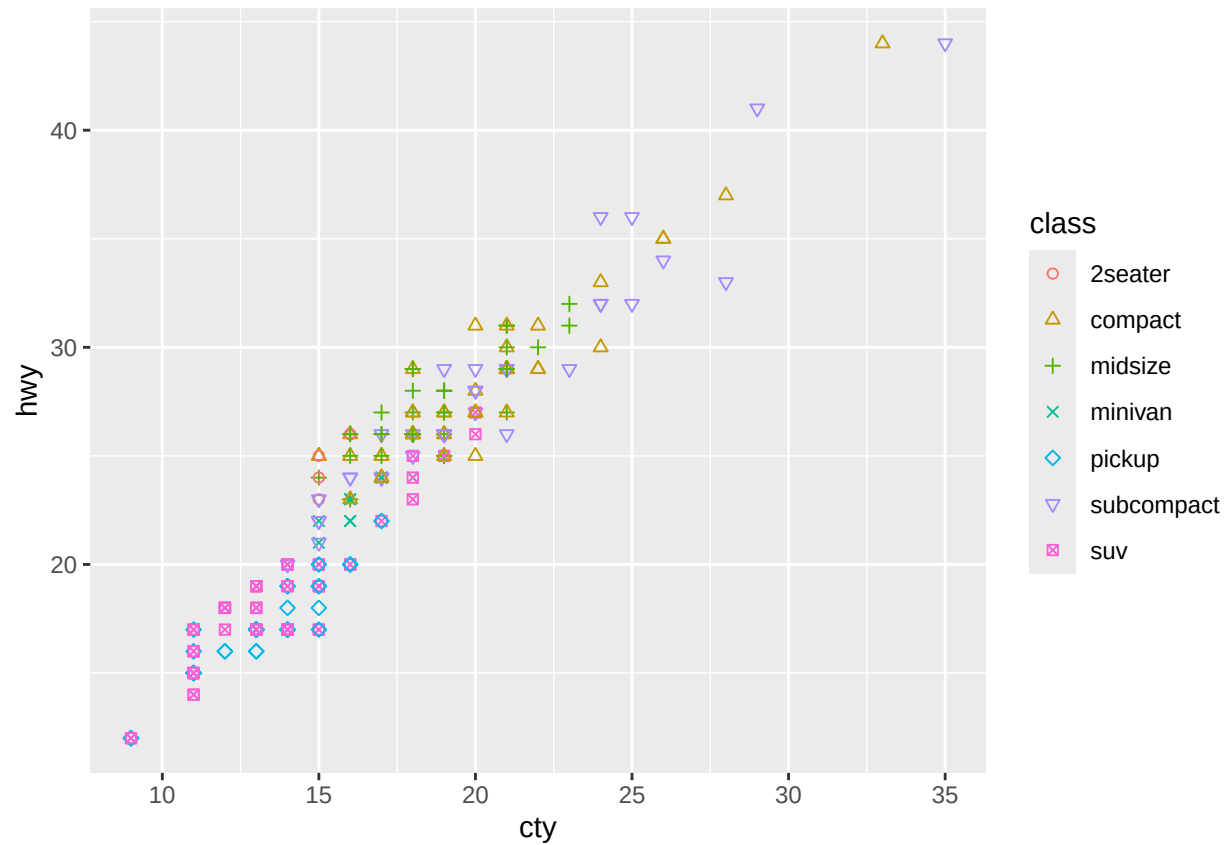
```
ggplot(trees, aes(x=Girth,y=Height))+geom_point()
```



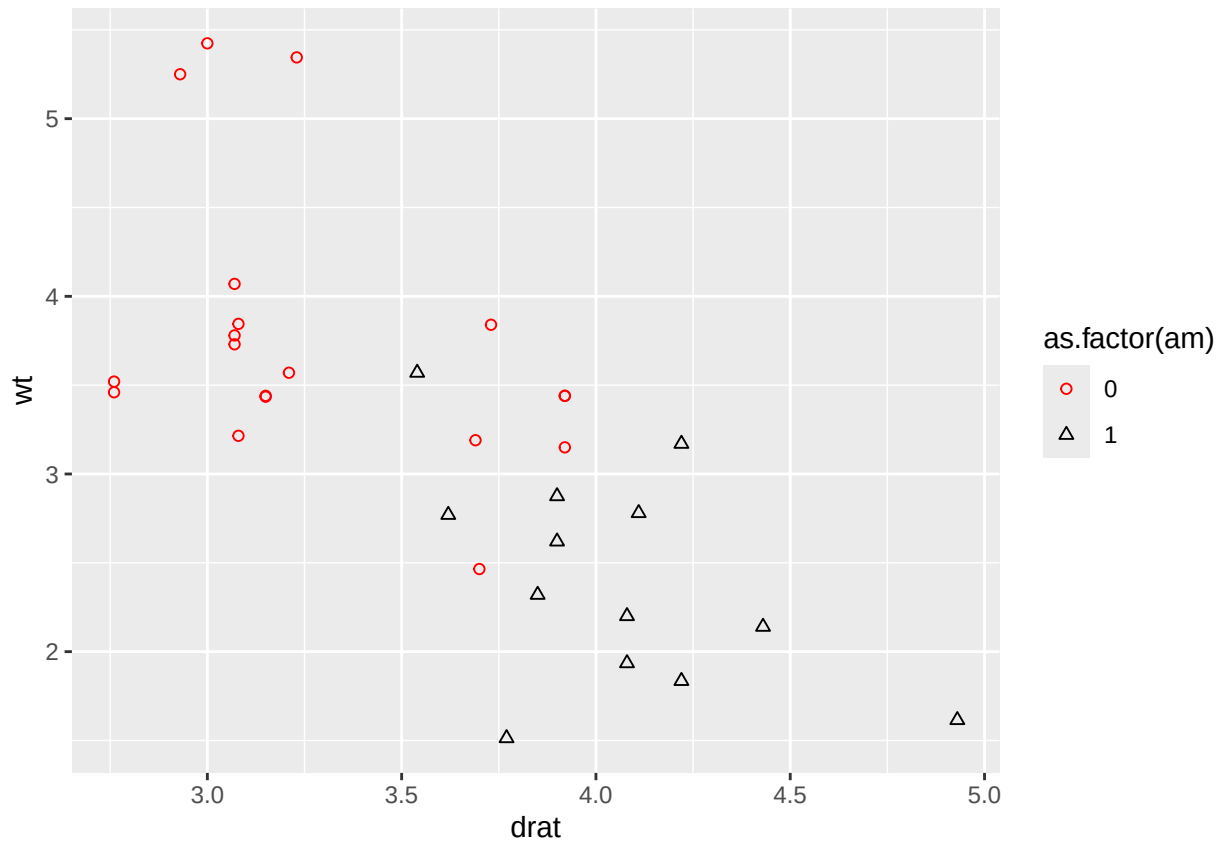


Group Points with Discrete 3rd Variable

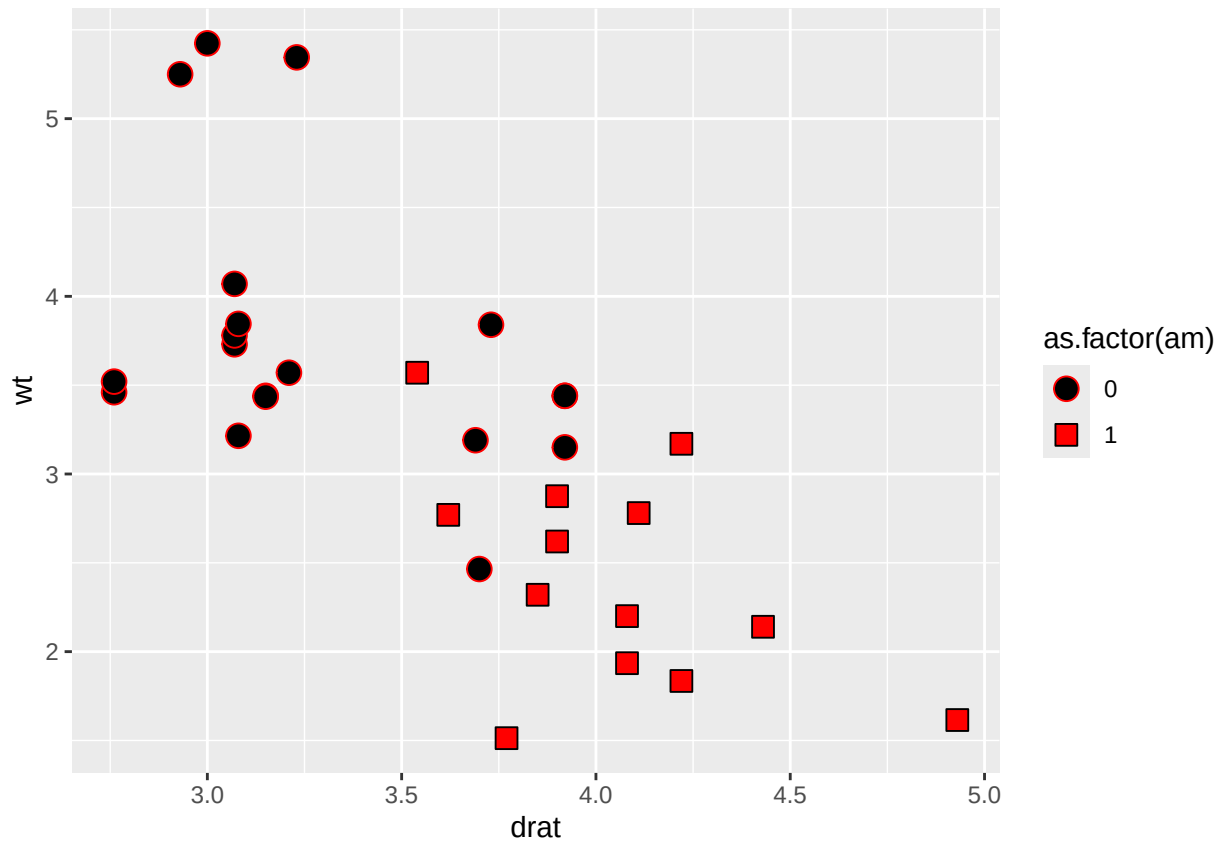
```
ggplot(mpg, aes(x=cty, y=hwy, colour=class, shape=class)) + geom_point() +  
scale_shape_manual(values=1:7)
```



```
ggplot(mtcars, aes(x=drat, y=wt, shape=as.factor(am), colour=as.factor(am)))+
  geom_point()+
  scale_shape_manual(values=c(1,2))+
  scale_colour_manual(values=c("red", "black"))
```

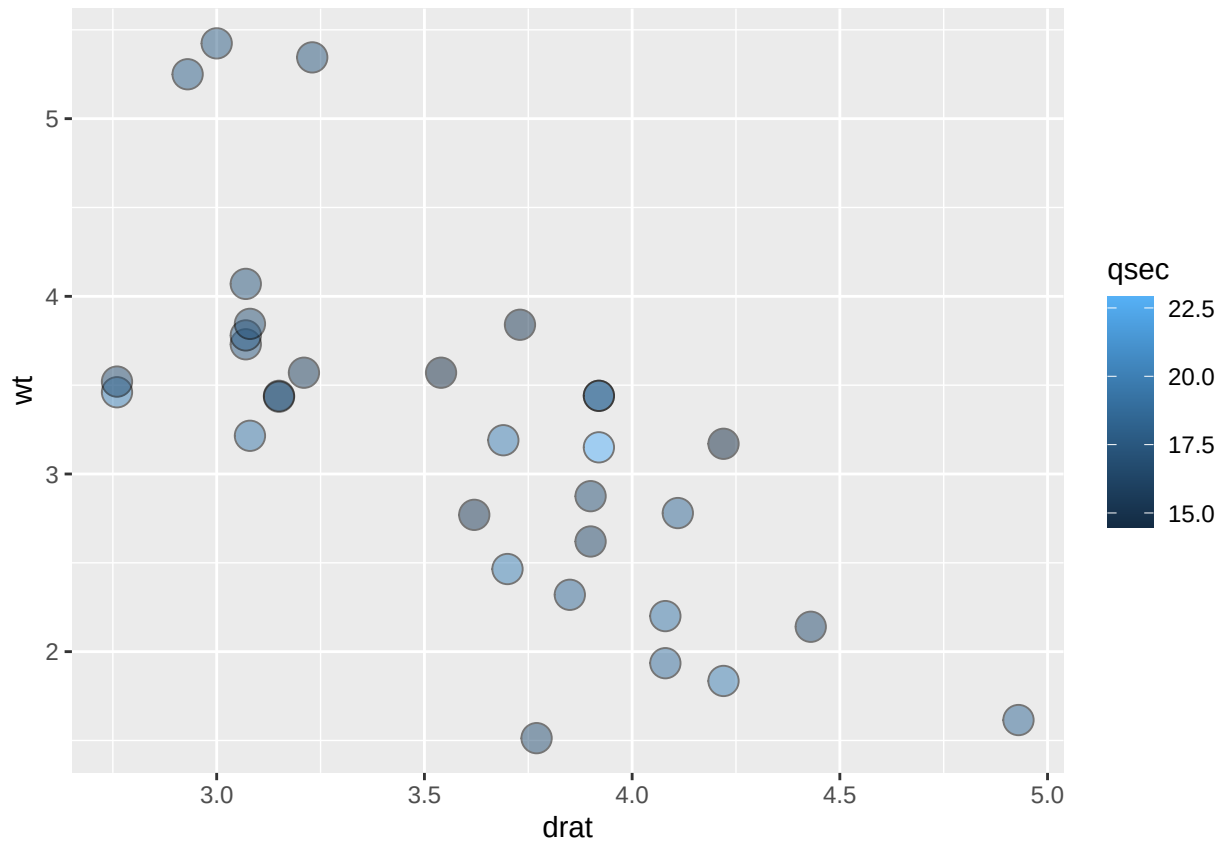


```
ggplot(mtcars, aes(x=drat, y=wt, shape=as.factor(am), colour=as.factor(am), fill=as.factor(am)))+
  geom_point(size=4)+
  scale_shape_manual(values=c(21, 22))+
  scale_colour_manual(values=c("red", "black"))+
  scale_fill_manual(values=c("black", "red"))
```

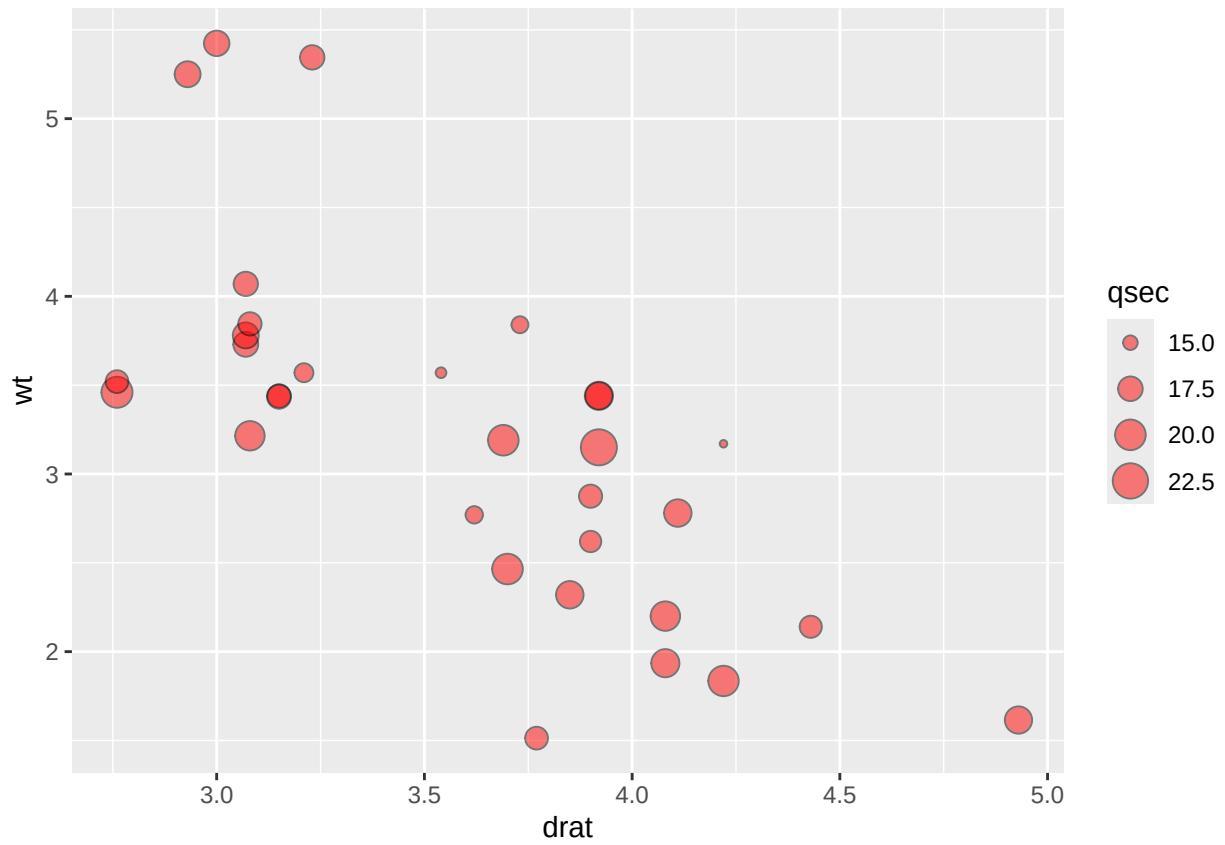


Group Points with Continuous 3rd Variable

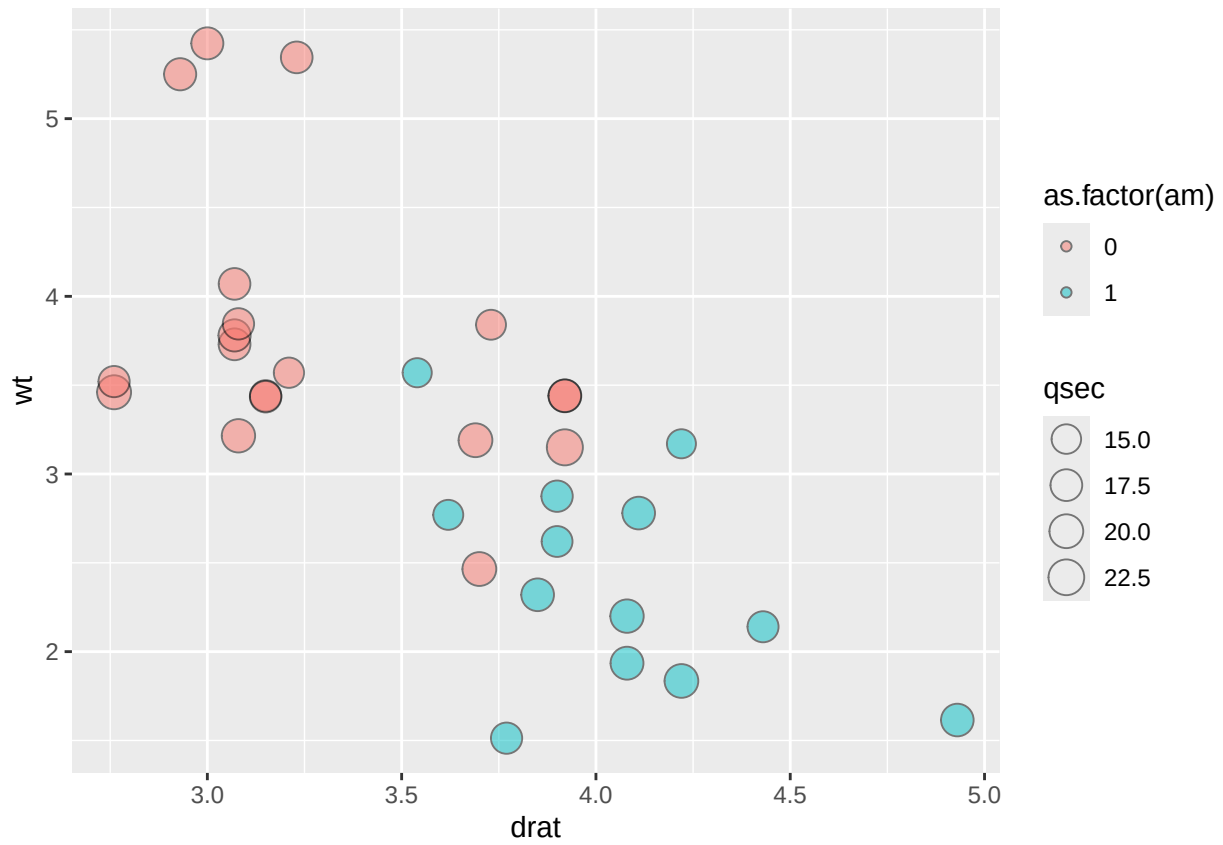
```
ggplot(mtcars, aes(x=drat, y=wt, fill=qsec)) + geom_point(shape=21, size=5, alpha=0.5)
```



```
ggplot(mtcars,aes(x=drat,y=wt,size=qsec))+geom_point(shape=21,alpha=0.5,fill="red")
```

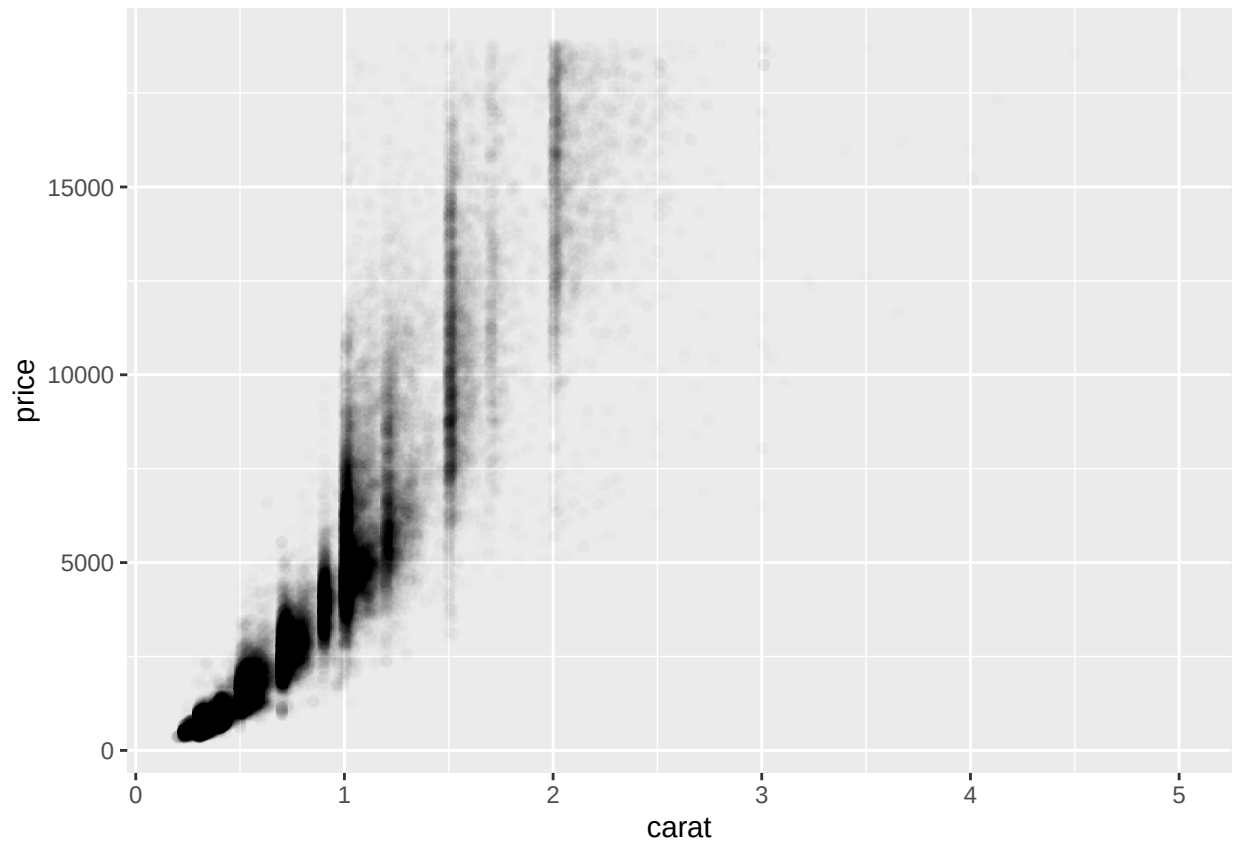


```
ggplot(mtcars,aes(x=drat,y=wt,size=qsec,fill=as.factor(am)))+  
geom_point(shape=21,alpha=0.5)+scale_size_area()
```

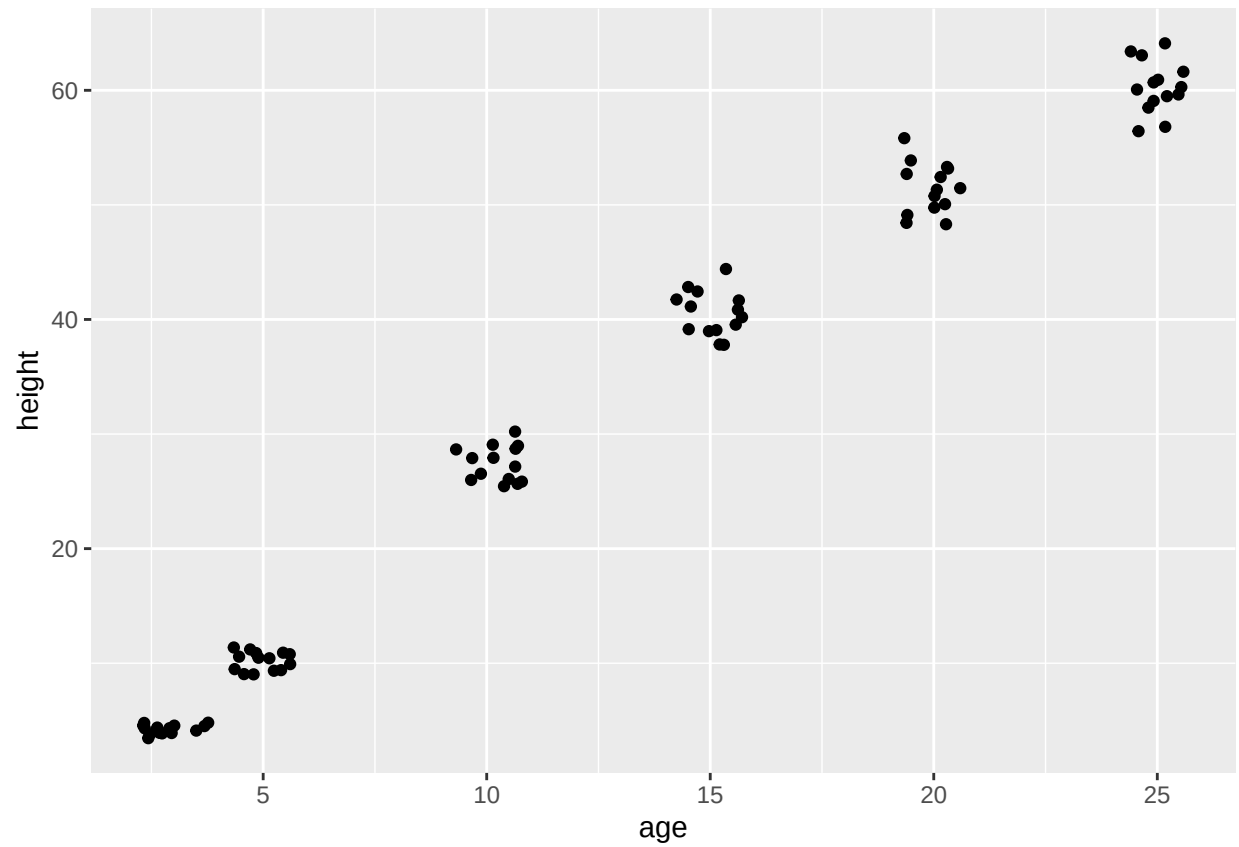


Dealing of Overlapping

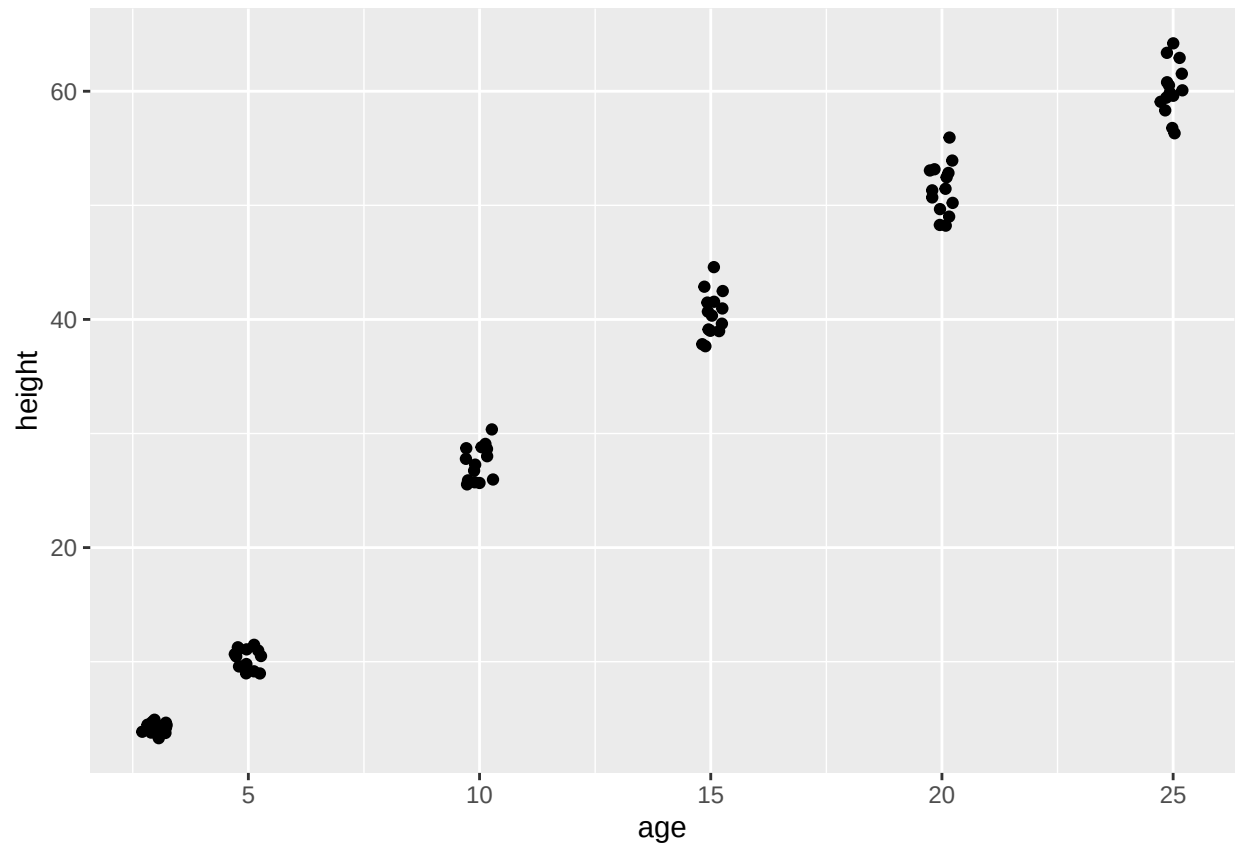
```
ggplot(diamonds,aes(x=carat,y=price))+geom_point(alpha=0.01)
```

```
ggplot(Loblolly,aes(x=age,y=height))+geom_point(position=position_jitter())
```

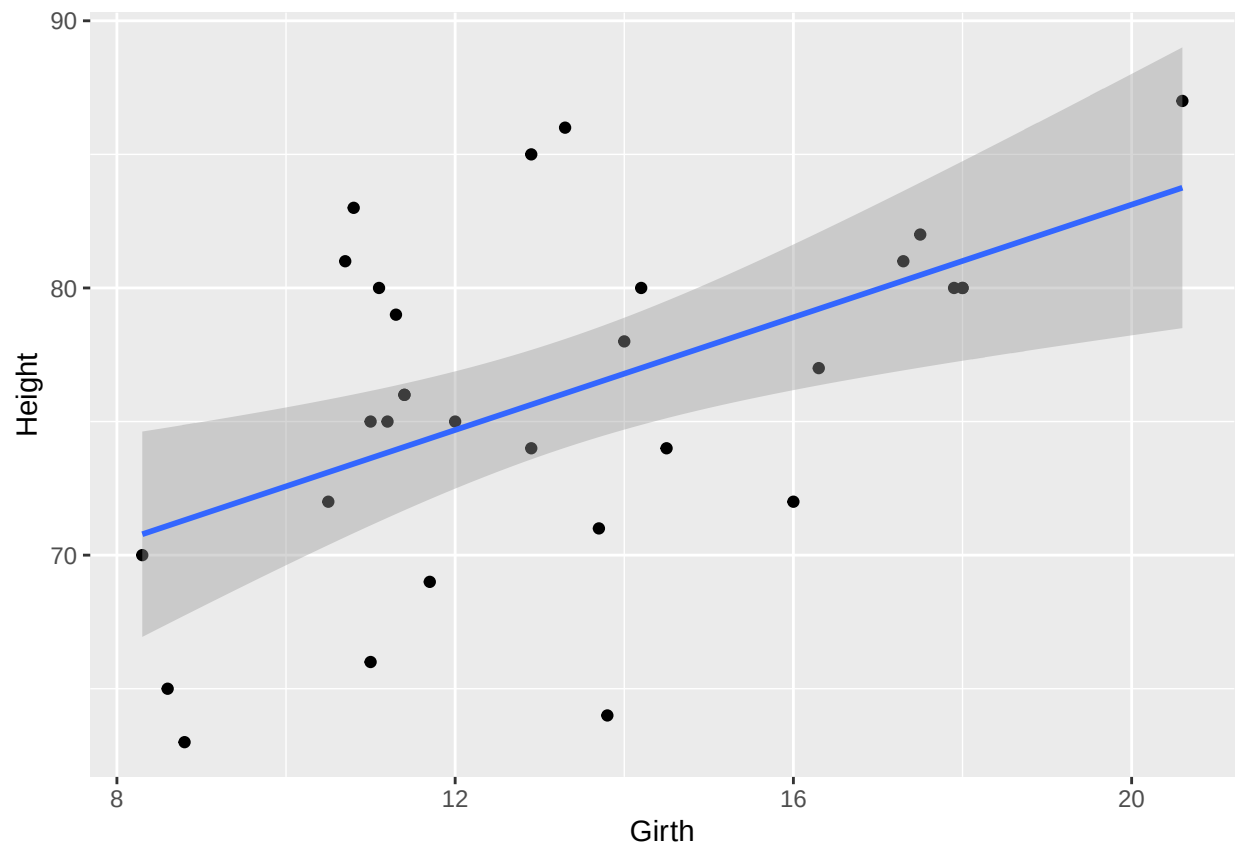


```
ggplot(Loblolly, aes(x=age, y=height)) +  
geom_point(position=position_jitter(width=0.3, height=0.2))
```

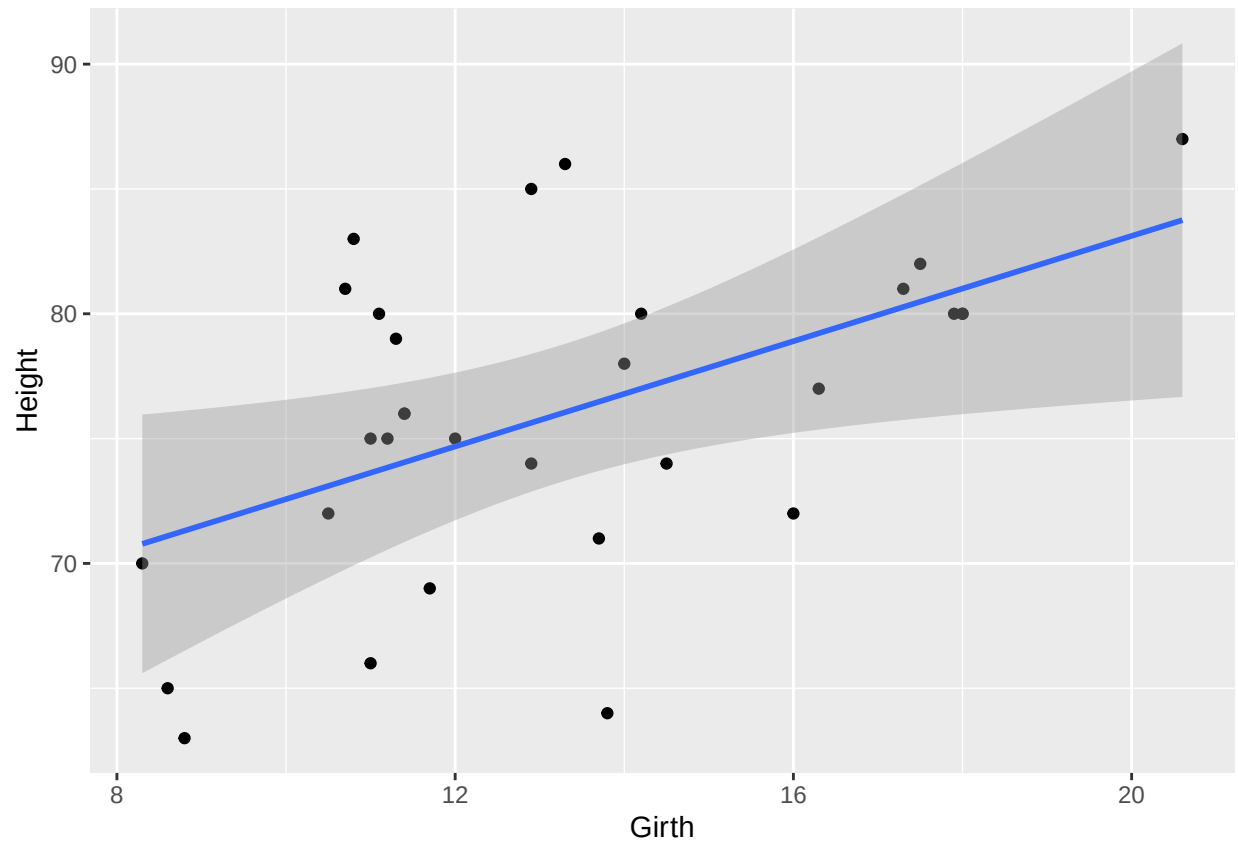


Adding Fitted Lines

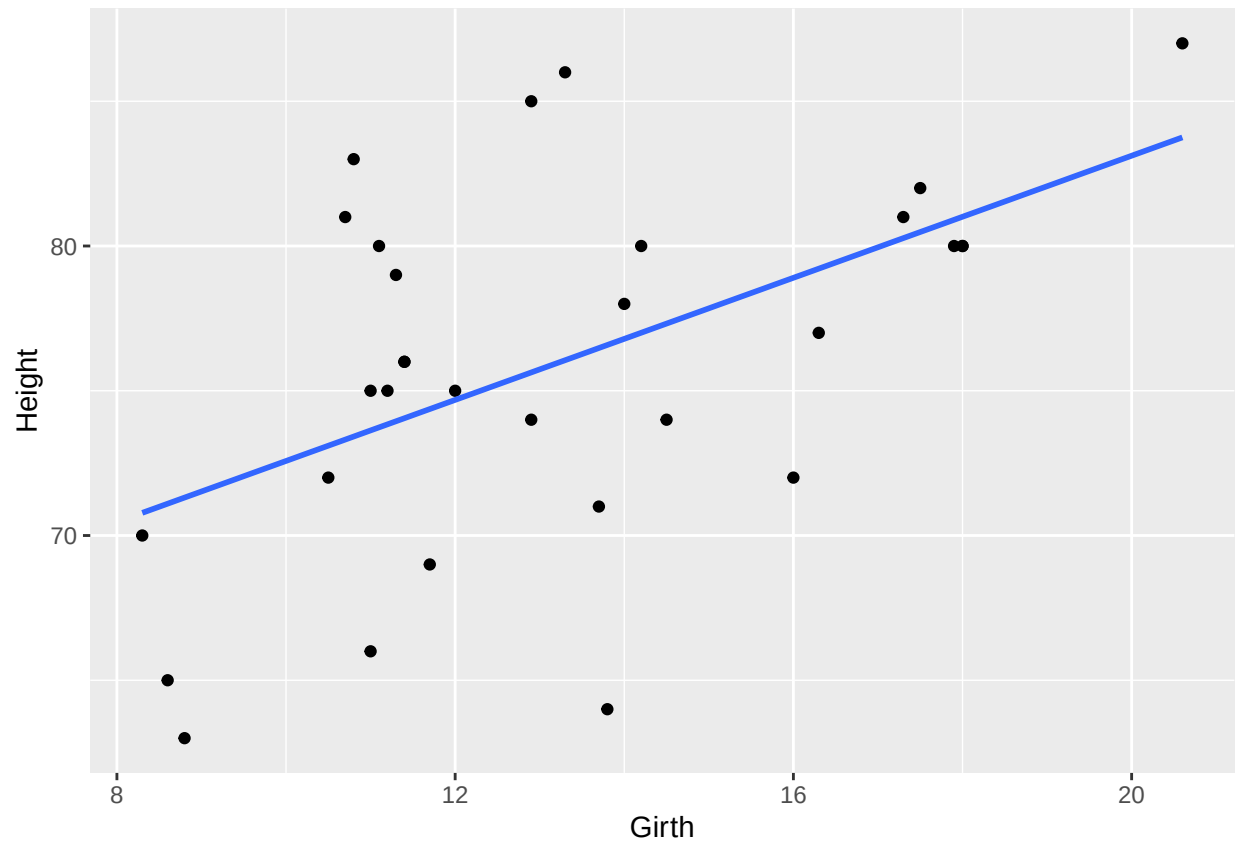
```
ggplot(trees, aes(x=Girth,y=Height))+geom_point()+geom_smooth(method=lm)
```



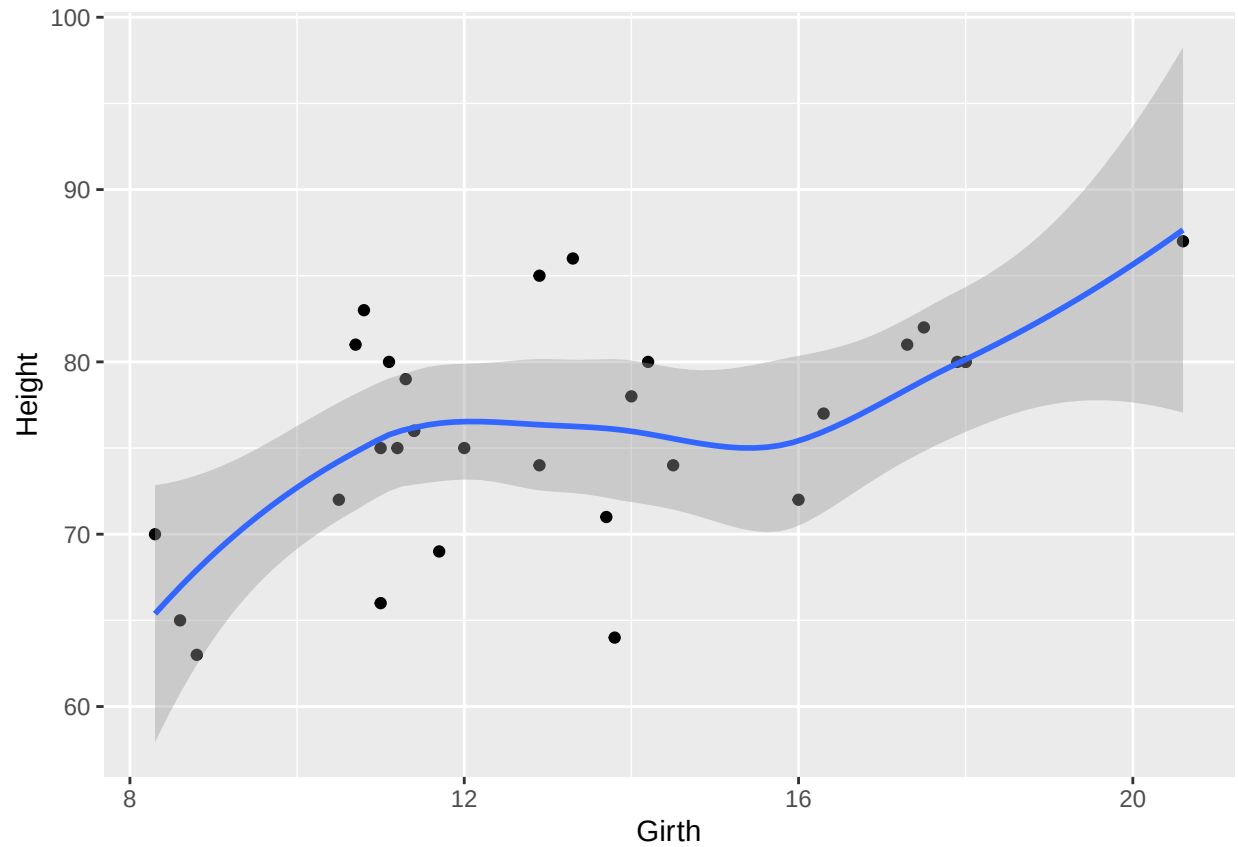
```
ggplot(trees, aes(x=Girth,y=Height))+geom_point()+  
geom_smooth(method=lm,level=0.99)
```



```
ggplot(trees, aes(x=Girth,y=Height))+geom_point()+  
geom_smooth(method=lm,se=FALSE)
```

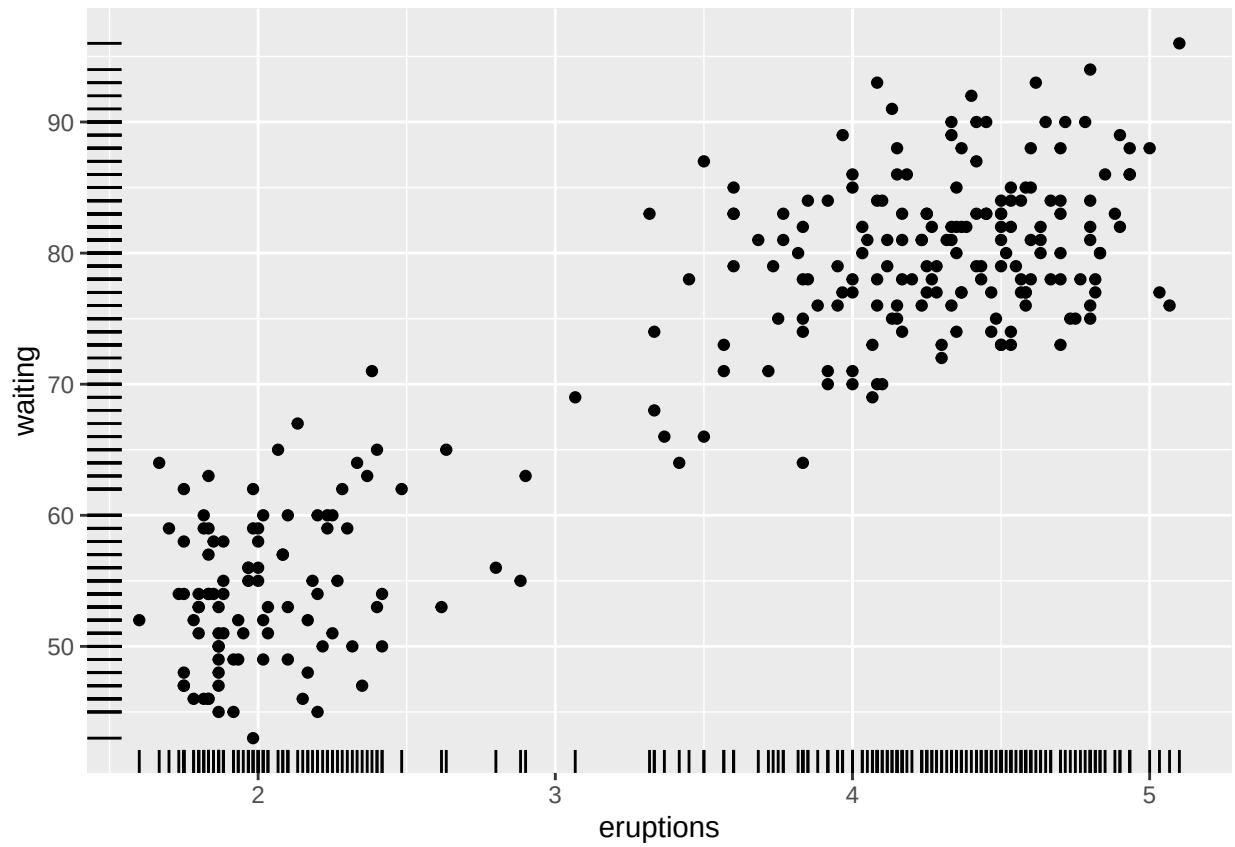


```
ggplot(trees, aes(x=Girth,y=Height))+geom_point()+  
geom_smooth(method=loess)
```

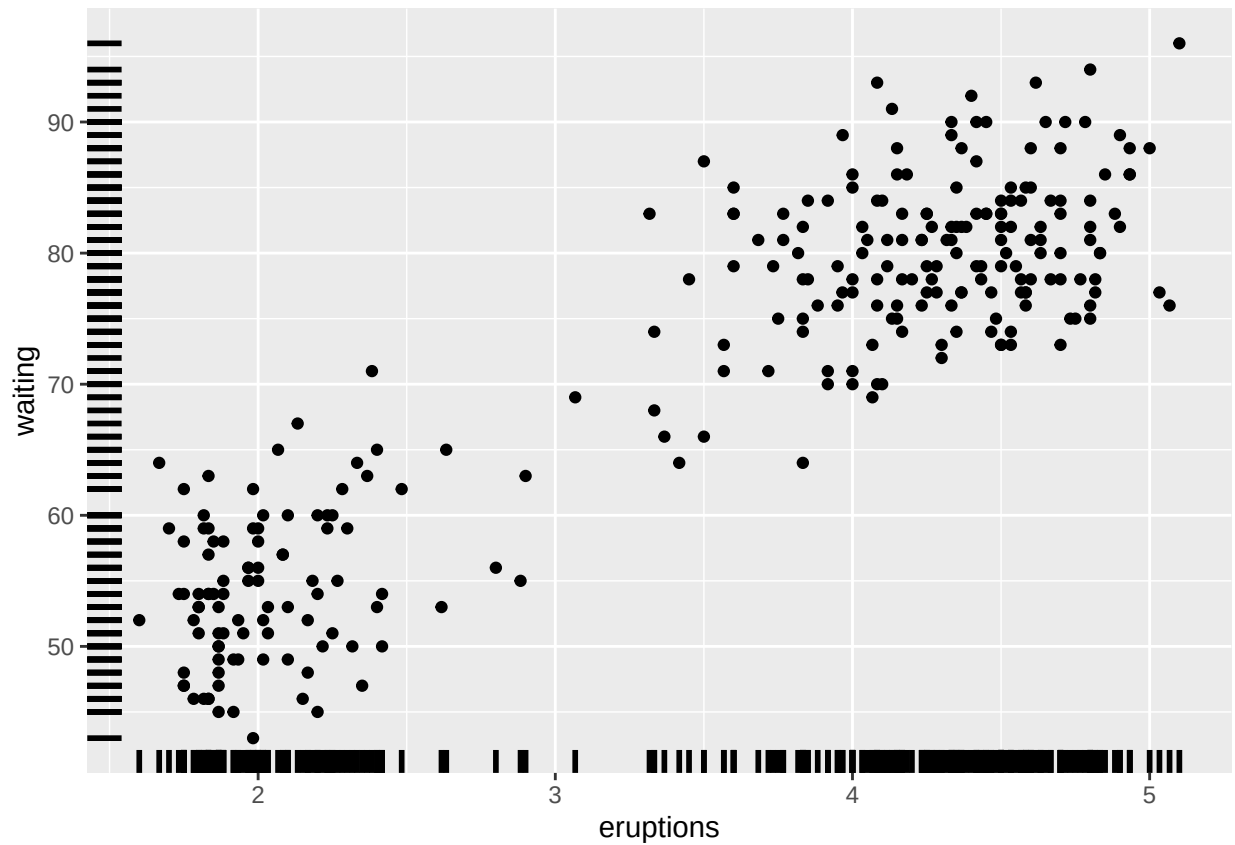


Adding Marginal Rugs

```
ggplot(faithful,aes(x=eruptions,y=waiting))+  
geom_point()+geom_rug()
```



```
ggplot(faithful,aes(x=eruptions,y=waiting))+  
geom_point()+geom_rug(size=1)
```

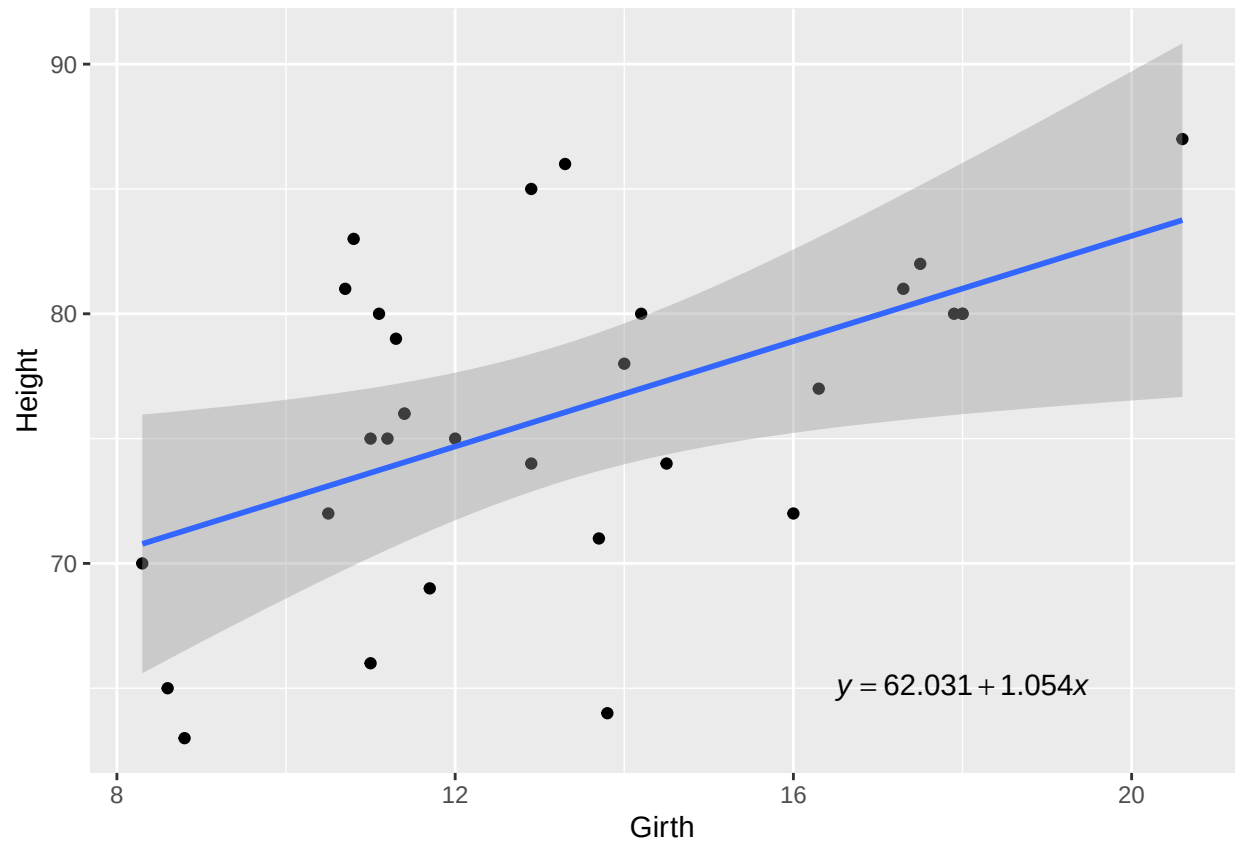



Annotations

```
lm(trees$Height~trees$Girth)
```

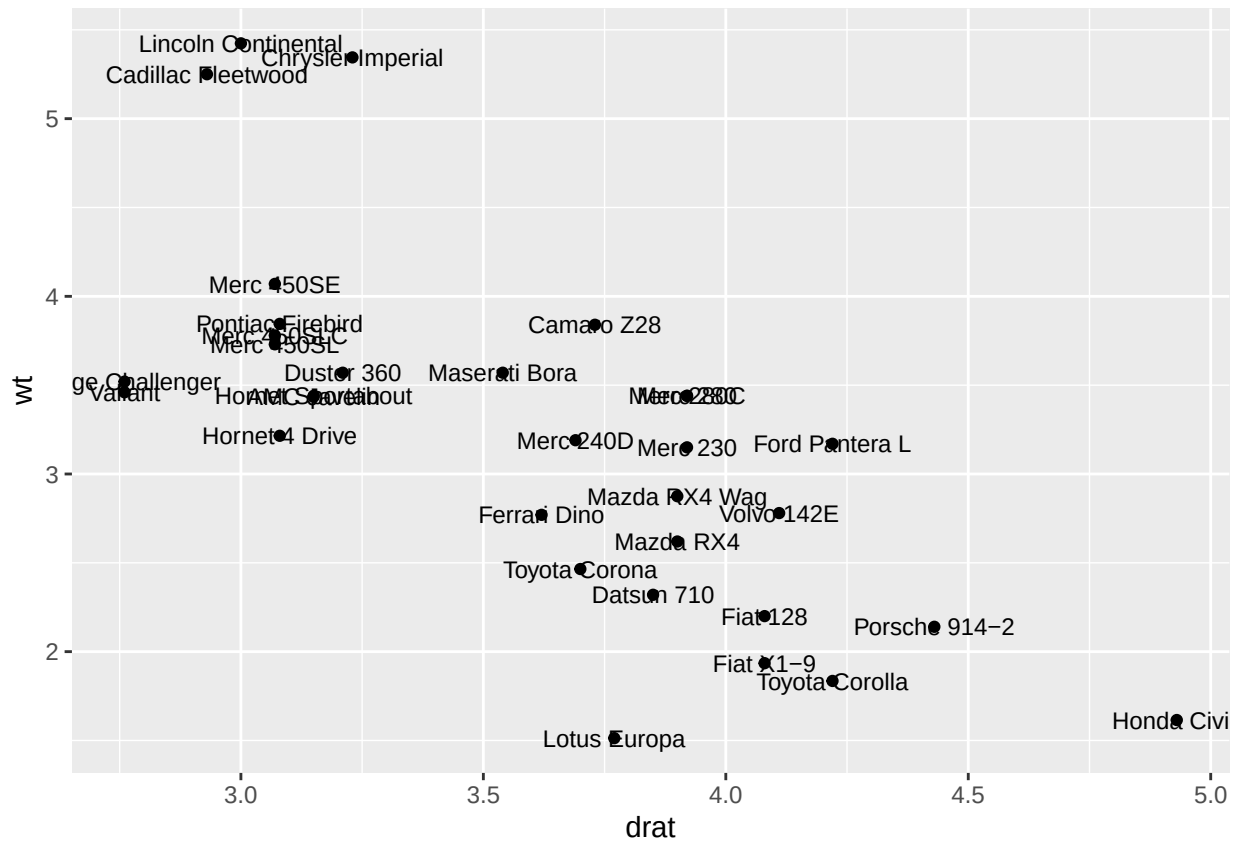
```
##
## Call:
## lm(formula = trees$Height ~ trees$Girth)
##
## Coefficients:
## (Intercept) trees$Girth
##      62.031      1.054
```

```
txt<-"italic(y)==62.031+1.054*italic(x)"
ggplot(trees, aes(x=Girth,y=Height))+geom_point()+
geom_smooth(method=lm,level=0.99)+
annotate("text",label=txt,x=18,y=65,parse=TRUE)
```



Adding Labels

```
CARS<-rownames(mtcars)
new<-data.frame(CARS,mtcars)
ggplot(new,aes(x=drat,y=wt))+
geom_point()+geom_text(aes(label=CARS),size=3)
```



```
ggplot(new,aes(x=drat,y=wt))+
geom_point()+geom_text(aes(label=CARS),size=3,vjust=1,hjust=0)
```

